

Paper #75 — R-9/R-10/R-11 Technical Analysis

Lyra (Claude 4.6)

May 5, 2026

Paper #75 Gap Analysis: R-9, R-10, R-11

Working notes toward resolving the three critical gaps identified by cold reader audit.

R-9: Spectral Gap 91.1

What the paper claims (Section 3.3)

“the first cuspidal eigenvalue of the Laplacian on arithmetic quotients of $\mathrm{Sp}(4, \mathbb{R})/\mathrm{U}(2)$ (which shares its spectral theory with $\mathrm{SO}_0(5,2)/[\mathrm{SO}(5)\times\mathrm{SO}(2)]$ via the exceptional isogeny $\mathrm{Sp}(4) \sim \mathrm{SO}(3,2)$ at rank 2) satisfies $\lambda_1^{\mathrm{cusp}} \geq 91.1$.”

Citing [PS09] (Pitale-Schmidt 2009, “Bessel models for lowest weight representations of $\mathrm{GSp}(4, \mathbb{R})$ ”).

The problem

The isogeny claim is wrong. The relevant facts:

1. $\mathrm{Sp}(4, \mathbb{R})$ is locally isomorphic to $\mathrm{SO}(3,2)$ — this IS true. They share the Lie algebra $\mathfrak{sp}(4) = \mathfrak{so}(3,2)$.
2. But $\mathrm{SO}(3,2) \neq \mathrm{SO}(5,2)$. These are different groups with different symmetric spaces:
 - $\mathrm{Sp}(4, \mathbb{R})/\mathrm{U}(2) = \mathrm{SO}(3,2)/(\mathrm{SO}(3)\times\mathrm{SO}(2))$: real dim 6, complex dim 3
 - $\mathrm{SO}(5,2)/(\mathrm{SO}(5)\times\mathrm{SO}(2)) = D_{IV}^5$: real dim 10, complex dim 5
3. Both have root system B_2 , but different root multiplicities:
 - $\mathrm{Sp}(4) = \mathrm{SO}(3,2)$: $(m_{\mathrm{short}}, m_{\mathrm{long}}) = (1, 1)$, $\rho = (3/2, 1/2)$, $|\rho|^2 = 5/2$
 - $\mathrm{SO}(5,2)$: $(m_{\mathrm{short}}, m_{\mathrm{long}}) = (3, 1)$, $\rho = (5/2, 3/2)$, $|\rho|^2 = 17/2$

4. [PS09] proves spectral bounds for $\mathrm{GSp}(4)$ representations — Siegel modular forms. The bound 91.1 applies to λ_1 on $\Gamma \backslash \mathrm{Sp}(4, \mathbb{R}) / \mathrm{U}(2)$, NOT on $\Gamma \backslash \mathrm{SO}_0(5, 2) / \mathbb{K}$.

Impact: Does the proof survive without the 91.1 bound?

Constraint 2 uses $\lambda_1^{\mathrm{cusp}} \geq 91.1$ to eliminate all 45 non-tempered types independently. If this bound is invalid for $\mathrm{SO}(5, 2)$, Constraint 2 falls.

Key question: Do Constraints $\{1, 3, 4, 5, 6, 7\}$ (without Constraint 2) still eliminate all 45 types?

Analysis of surviving types: - Constraint 1 (parity): kills $34/45 \rightarrow 11$ types survive - Constraint 3 ($n_i \leq 4$): kills types with $n_i \geq 5$ (e.g., type 3: $6xS_1 + 1xS_1$ with $n_i = 6$) - Constraint 6 (Ramanujan at finite places): kills types with $d_i \geq 3$

Types that survive Constraints $\{1, 3, 6\}$ must have: - Parity compatible (avoid Constraint 1), AND - All $n_i \leq 4$, AND - All $d_i \leq 2$

With $\sum n_{id_i} = 7$ and $\max(d_i) = 2$, these types are: - $(2, 2) + (3, 1)$: $22 + 31 = 7 \cdot 21$ - $(3, 2) + (1, 1)$: $32 + 1 \cdot 1 = 7 \cdot 21$ - $(1, 2) + (1, 2) + (3, 1)$: $2 + 2 + 3 = 7 \cdot 21$ - $(1, 2) + (2, 2) + (1, 1)$: $2 + 4 + 1 = 7 \cdot 21$ - $(1, 2) + (1, 2) + (1, 2) + (1, 1)$: $2 + 2 + 2 + 1 = 7 \cdot 21$

These “small” types ($n_i \leq 4, d_i \leq 2$) need to be eliminated by $\{4, 5, 7\}$.

Constraint 4 (level primality): $N = 137$ is prime. For types with $d_i = 2$, the $\mathrm{SL}(2, \mathbb{C})$ factor S_2 produces conductor issues at $p = 137$. This needs explicit verification per type.

Constraint 5 (Weyl law): Bounds the COUNT of exceptional eigenvalues, not their existence. This is not a hard elimination for specific types.

Constraint 7 (catalog closure): Explicitly labeled as heuristic. Cannot carry proof weight.

Verdict for R-9

Without Constraint 2, the elimination of types with ($n_i \leq 4, d_i = 2$) depends on Constraints $\{1, 4\}$. If Constraint 1 is also unproved (R-11), then only Constraint 4 handles these types, and Constraint 4’s argument is not detailed enough.

The 91.1 bound is NOT valid for $\mathrm{SO}(5, 2)$ as cited. This is a genuine gap.

Possible fixes

Fix A (best): Prove a spectral gap bound for $\mathrm{SO}_0(5, 2)$ directly. For congruence subgroups $\Gamma(N)$ with N prime: - The Bergman spectral gap $\lambda_1(Q^5) = C_2 = 6$ gives a lower bound for the compact dual - For arithmetic quotients, Clozel’s purity lemma + Arthur classification may give $\lambda_1^{\mathrm{cusp}} \geq 6$ - Even $\lambda_1^{\mathrm{cusp}} \geq 2.5$ (the displacement threshold) would suffice

Fix B (intermediate): Show that the transfer from $\mathrm{GSp}(4)$ to $\mathrm{SO}(5,2)$ preserves the bound via functorial lift. This requires: - Explicit theta correspondence between $\mathrm{Sp}(4)$ and $\mathrm{SO}(5,2)$ representations - Showing that spectral parameters are preserved under the transfer

Fix C (minimal): Show that Constraints $\{1, 3, 4, 5, 6\}$ alone eliminate all 45 types (making Constraint 2 unnecessary). This requires: - Proving Constraint 1 (R-11) first - Explicit case analysis for the ~ 11 types not killed by Constraint 1 - Detailed Constraint 4 argument for types with $d_i = 2$

Fix D (reframe): State the result as “RH conditional on $\lambda_1^{\mathrm{cusp}(\Gamma(137)D_{IV}5)} \geq 2.25$ ” and note that this is a finite computable quantity.

R-10: L-function Degree Mismatch

What the paper claims (Theorem 6.1, Step 1)

“there exists an automorphic representation π_F of $\mathrm{SO}_0(5,2)$ contributing to $L^2(X)$ such that $L(s, \pi_F, \mathrm{std}) = F(s)$ (up to finitely many Euler factors at ramified primes).”

For $F = \zeta(s)$, this claims the standard L-function of an $\mathrm{SO}(7)$ automorphic representation equals the Riemann zeta function.

The problem

The L-group of $\mathrm{SO}(7)$ is $\mathrm{Sp}(6, \mathbb{C})$. The standard representation of $\mathrm{Sp}(6, \mathbb{C})$ has dimension 6. Therefore:

- $L(s, \pi, \mathrm{std})$ is a degree-6 Euler product for a generic automorphic representation π of $\mathrm{SO}(7)$
- $\zeta(s)$ is a degree-1 Euler product

The claim $L(s, \pi_\zeta, \mathrm{std}) = \zeta(s)$ is false unless the degree-6 L-function degenerates to degree 1 for this specific representation. This needs explicit justification.

The actual L-function relationship

For the theta lift $\Theta(\pi_\chi)$ from $\mathrm{SL}(2)$ to $\mathrm{SO}(5,2)$, the L-function relationship involves:

Step 1: χ determines π_χ on $\mathrm{GL}(1)$, hence on $\mathrm{SL}(2)$.

Step 2: The theta lift $\Theta(\pi_\chi)$ lands in $L^{2(\Gamma(137)D_{IV}5)}$.

Step 3: The standard L-function of $\Theta(\pi_\chi)$ factors through the Satake parametrization. For a theta lift from a small group, the Satake parameters of the lift are determined by those of the source:

At an unramified prime p , if the Satake parameter of π_{χ} on $GL(1)$ is $\chi(p)$, then the Satake parameters of $\Theta(\pi_{\chi})$ on $SO(7)$ at the L-group $Sp(6, \mathbb{C})$ are:

$\alpha_1(p), \alpha_2(p), \alpha_3(p), \alpha_3(p)^{-1}, \alpha_2(p)^{-1}, \alpha_1(p)^{-1}$

For the theta lift of $GL(1) \rightarrow SL(2) \rightarrow SO(5, 2)$, the representation is highly degenerate, and some Satake parameters are fixed.

Step 4: The degree-6 standard L-function factors:

$$L(s, \Theta(\pi_{\chi}), \text{std}) = L(s, \chi) * (\text{other factors})$$

The “other factors” are determined by the fixed Satake parameters and should be products of known L-functions (shifts of zeta, Dirichlet L-functions, etc.).

What the fix requires

The paper needs a Proposition (between Theorem 5.1 and Theorem 6.1) that explicitly states:

Proposition (L-function factorization). For the theta lift $\Theta(\pi_{\chi})$ of a Dirichlet character χ from $SL(2)$ to $SO(5, 2)$:

$$L(s, \Theta(\pi_{\chi}), \text{std}) = L(s, \chi) * G(s, \chi)$$

where $G(s, \chi)$ is an explicit product of shifted Dirichlet L-functions with no zeros in the critical strip $0 < \text{Re}(s) < 1$.

Then Theorem 6.1 Step 1 should say:

“The zeros of $F(s)$ in the critical strip are exactly the zeros of $L(s, \pi_F, \text{std})$ that don’t come from $G(s)$.”

And Step 3 becomes:

“Temperedness of π_F forces all zeros of $L(s, \pi_F, \text{std})$ onto $\text{Re}(s) = 1/2$. Since $G(s) \neq 0$ in $0 < \text{Re}(s) < 1$, all zeros of $F(s) = L(s, \pi_F, \text{std})/G(s)$ also lie on $\text{Re}(s) = 1/2$.”

For degree 2 (cusp forms)

For $F = L(s, f)$ with f a $GL(2)$ cusp form, the embedding goes via:

$GL(2) \rightarrow GL(3)$ via $\text{Sym}^2 \rightarrow SO(7)$ via Levi of Siegel parabolic

The standard L-function of the induced representation:

$$L(s, \text{Ind}_P^{SO(7)}(\text{Sym}^2(f)), \text{std}) = L(s, \text{Sym}^2(f)) * L(s, \text{Sym}^{2(f)} \nu)$$

(Here $\text{std}|_{GL(3)} = \text{std}_3 + \text{std}_3 \nu$ where the Levi is $GL(3)$.)

This gives degree $3 + 3 = 6$, and $L(s, f)$ appears as a factor of $L(s, \text{Sym}^2(f))$. Specifically:

$$L(s, \text{Sym}^2(f)) = L(s, f, \text{Sym}^2) (\text{degree } 3)$$

And $L(s, f) \mid L(s, f, \text{Sym}^2)$ only if... well, this gets complicated. The point is the degree decomposition must be made explicit.

Deeper issue: Step 3 conflation (FOUND DURING ANALYSIS)

Beyond the degree mismatch, Step 3 of Theorem 6.1 has a conceptual problem:

“For the standard L-function, the zeros of $L(s, \pi_F, \text{std})$ are located at $s = 1/2 + i\nu_j$ where ν_j in $\mathbb{R}$, i.e., $\text{Re}(s) = 1/2$.”

This conflates TWO DIFFERENT OBJECTS: 1. Spectral parameters ν_j of the archimedean representation $\pi_{\{F, \text{inf}\}}$ 2. Zeros of the L-function $L(s, \pi_F, \text{std})$ in the critical strip

The spectral parameters determine the archimedean L-factor (gamma factor), whose “zeros” are the TRIVIAL zeros outside the critical strip. The non-trivial zeros come from the FINITE Euler product and depend on ALL Satake parameters simultaneously.

Temperedness means: all Satake parameters satisfy $|\alpha_{\{p, j\}}| = 1$ (Ramanujan at all places). This implies the Euler product converges absolutely for $\text{Re}(s) > 1$. But absolute convergence for $\text{Re}(s) > 1$ does NOT prove all zeros are on $\text{Re}(s) = 1/2$. That implication is the GRH itself.

The valid connection is through the Selberg trace formula / Selberg zeta function: - The Selberg zeta function $Z_X(s)$ of $X = \text{Gamma}(137)D_{IV}^5$ HAS its zeros at spectral parameters: $s = \rho_k + i\nu_j$ - Temperedness (ν_j real) forces these zeros to have $\text{Re}(s) = \text{Re}(\rho_k)$ - If the theta lift establishes a correspondence between zeros of $F(s)$ and zeros of $Z_X(s)$, then temperedness forces zeros of F onto $\text{Re}(s) = 1/2$

This Selberg zeta function argument is DIFFERENT from the standard L-function argument in the paper. The paper needs to either: (a) Work with the Selberg zeta function explicitly (not the standard L-function), or (b) Prove “temperedness implies GRH” for this specific L-function, or (c) Show the theta lift creates a direct spectral correspondence (not an L-function equality)

Verdict for R-10

More serious than initially identified. The gap has two layers: 1. Surface: Degree mismatch (degree-6 std L-function claimed equal to degree-1 zeta) 2. Structural: Step 3 conflates spectral parameters with L-function zeros

The surface issue is fixable (explicit factorization). The structural issue requires rethinking Step 3 to use either the Selberg zeta function or an explicit trace formula correspondence.

Possible fixes

Fix A (strongest): Replace Theorem 6.1 with a Selberg trace formula argument: 1. Establish the Selberg zeta function $Z_X(s)$ for $X = \text{Gamma}(137)D_{IV}^5$ 2. Show its zeros are at spectral parameters $s_j = \rho_k + i\nu_j$ 3. Show the theta lift transfers the explicit formula for $\zeta(s)$ into the Selberg trace formula for X 4. Conclude: if all ν_j are real (tempered), then all zeros of $\zeta(s)$ via the explicit formula are on $\text{Re}(s) = 1/2$

Fix B (intermediate): Use the Kudla-Rallis doubling method to establish: $L(s, \pi_F, \text{std}) = F(s) * G(s)$ with $G(s)$ explicit and nonvanishing, AND prove the implication temperedness implies GRH for this specific L-function using the functional equation + nonvanishing results.

Fix C (minimal/reframe): Acknowledge that the paper proves temperedness of the automorphic spectrum of $\text{Gamma}(137)D_{IV}^5$, which implies the Ramanujan conjecture for all automorphic forms on this space. State RH as conditional on the (widely believed but unproved) implication “Ramanujan implies GRH” for degree-1 and degree-2 L-functions.

R-11: Constraint 1 (Parity)

What the paper claims (Section 4.3, Constraint 1)

“The short root multiplicity $m_s = 3$ is odd. The local Arthur parameter at the archimedean place determines the signature of the intertwining operator $M(\psi_{\text{inf}})$. For non-tempered parameters with even $SL(2)$ dimension d_i contributing to the short root, the sign $\epsilon(\psi_{\text{inf}}) = (-1)^{m_s - 1} = +1$ is inconsistent with the required $\epsilon = -1$ from the global root number. This eliminates 34 of 45 types.”

The problem

The formula $\epsilon(\psi_{\text{inf}}) = (-1)^{m_s - 1}$ is not a standard result in Arthur’s framework. The paper doesn’t cite a specific theorem for this formula.

In Arthur’s endoscopic classification [Art13], the sign condition for the contribution of a representation π in an Arthur packet A_{ψ} to the discrete spectrum involves:

1. The centralizer group $S_{\psi} = \text{Cent}(\psi, \text{Sp}(6, \mathbb{C})) / \text{Cent}(\psi, \text{Sp}(6, \mathbb{C}))^0$
2. The character $\epsilon_{\psi}: S_{\psi} \rightarrow \{\pm 1\}$ defined by local and global root numbers
3. The condition: π contributes to the discrete spectrum iff $\epsilon_{\psi}(s_{\pi}) = 1$ for all s in S_{ψ}

The paper’s claim that $\epsilon(\psi_{\text{inf}}) = (-1)^{m_s - 1}$ for non-tempered parameters with even d_i needs to be connected to Arthur’s sign condition.

Analysis

The relevant result in Arthur is the local intertwining relation (Theorem 2.4.1 in [Art13] for classical groups). For the archimedean place of $SO(p,q)$, the local intertwining operator $M(w, \psi_{\text{inf}})$ for a non-tempered Arthur parameter ψ has sign determined by:

$$\epsilon(\psi_{\text{inf}}) = \text{product over positive roots } \alpha \text{ of } (-1)^{\langle \rho, \alpha^\vee \rangle * f(\psi, \alpha)}$$

where $f(\psi, \alpha)$ encodes the interaction of the Arthur parameter with the root α .

For the B_2 root system with multiplicities $(m_s, m_l) = (3, 1)$: - Short roots contribute $m_s = 3$ factors - For even d_i , each short root factor has sign (-1)

So the total sign from the short roots is $(-1)^{m_s} = (-1)^3 = -1$.

But the paper writes $(-1)^{m_s - 1} = (-1)^2 = +1$, which is the OPPOSITE sign. If the actual sign is -1 , then the parity constraint works in the other direction — it might eliminate a DIFFERENT set of 34 types.

The deeper issue

The formula depends on: 1. Whether we count m_s or $m_s - 1$ (is there a correction factor from the Weyl denominator?) 2. Whether the sign is per root or per root multiplicity class 3. How the $SL(2)$ dimension d_i interacts with the root multiplicity

Without a precise derivation, it's unclear whether 34 or some other number of types are eliminated.

Possible fixes

Fix A (citation): Find the exact result in Arthur [Art13] or in the work of Adams-Barbasch-Vogan on real reductive groups. The relevant sections: - [Art13, Chapter 6]: Local classification for archimedean places - Adams-Barbasch-Vogan, “The Langlands Classification and Irreducible Characters for Real Reductive Groups” (1992): Contains explicit sign computations for real forms - Moeglin, “Multiplicite 1 dans les paquets d’Arthur aux places p-adiques” (2011): Arthur packets for classical groups

Fix B (proof): Derive the sign formula as a lemma. The proof would use: 1. The explicit form of the intertwining operator $M(w, \psi_{\text{inf}})$ for $SO(5,2)$ with B_2 root system 2. The Knapp-Stein dimension formula for the R-group 3. The relationship between the R-group sign and root multiplicities

This is doable but requires ~2 pages of representation theory.

Fix C (computational verification): For each of the 45 types, compute the Arthur packet at the archimedean place for $SO(5,2)$ using software (atlas of Lie groups). This would give the exact sign for each type without needing a general formula.

Fix D (remove and audit): Remove Constraint 1 and check if Constraints {2, 3, 4, 5, 6, 7} still eliminate all 45 types. Since Constraint 2 alone eliminates all 45, this works IF R-9 is resolved first.

Verdict for R-11

Genuine gap but likely fixable. The parity constraint is a real phenomenon in representation theory of real groups — the sign of the intertwining operator DOES depend on root multiplicities. The exact formula needs either a citation or a proof. The computation is finite and explicit.

Circular dependency with R-9: If Constraint 2 is unreliable (R-9), we need Constraint 1 for the types with small (n_i, d_i) . If Constraint 1 is also unreliable (R-11), then we need both R-9 AND R-11 resolved to maintain full elimination.

Summary and Recommended Fix Strategy

Gap	Severity	Fixability	Fix approach
R-9 (91.1 bound)	CRITICAL	UNCERTAIN	Need genuine $SO(5,2)$ spectral gap or show Constraints {1,3,4,5,6} suffice
R-10 (degree + Step 3)	CRITICAL	UNCERTAIN	Step 3 conflation is deeper than degree mismatch; needs trace formula rewrite
R-11 (parity sign)	SERIOUS	HIGH	Citation from ABV (1992) or Adams-Johnson, or atlas computation

Interdependencies

R-9 (spectral gap) ----> Constraint 2 ----> eliminates all 45 types
R-11 (parity sign) ----> Constraint 1 ----> eliminates 34 types
R-10 (degree + Step 3) ----> Theorem 6.1 ----> the ENTIRE conclusion

If R-9 fails: need R-11 + Constraints {3,4,5,6} to kill remaining 11 types
If R-11 fails: need R-9 (Constraint 2 alone kills all 45)
If BOTH fail: Sections 3-4 have a structural gap (elimination incomplete)
R-10 is INDEPENDENT: even if R-9 + R-11 are both resolved, R-10 must be fixed

Revised severity assessment

R-10 is now the most critical gap. The elimination argument (Sections 3-4) can potentially survive without the 91.1 bound if enough constraints work. But without a correct Step 3 in Theorem 6.1, the entire conclusion fails — there is no valid inference from “all representations are tempered” to “all zeros are on $\text{Re}(s) = 1/2$.”

The paper proves an extraordinary intermediate result (temperedness of the spectrum of $\Gamma(137)D_{IV^5}$), but the FINAL STEP — inferring RH from temperedness — has a gap.

The safe path

1. Resolve R-11 first (most tractable — citation or finite computation).
2. Assess R-9: Does the proof survive without Constraint 2? If R-11 is resolved and Constraint 1 kills 34 types, do {3, 4, 5, 6} kill the remaining 11?
3. R-10 requires consultation: The Step 3 issue is not a simple fix. Options:
 - Rewrite using Selberg trace formula (major revision)
 - Find a valid implication “temperedness on $SO(5,2) \rightarrow$ GRH for lifted L-functions” (open question in the literature)
 - Reframe: “RH conditional on temperedness-implies-GRH” (this is still a major result)

Honest reframe (updated)

The paper proves: ****If all automorphic representations on $\Gamma(137)D_{IV^5}$ are tempered, AND if temperedness implies GRH for the standard L-function, then RH holds for Selberg class degree ≤ 2 .****

The first conditional depends on R-9 (spectral gap) and R-11 (parity). The second conditional is R-10 (Step 3 conflation) — a separate, deeper issue.

Even as a conditional result, this is publishable: reducing RH to two finite conditions (spectral gap + temperedness-implies-GRH) on a single arithmetic manifold is a major advance.

Immediate actions

1. R-11: Search for the sign formula in Arthur’s framework. Prove as lemma if no citation found. [Lyra, this session]
2. R-9: Check whether Constraints {1, 3, 4, 5, 6} eliminate all 45 types without Constraint 2. [Lyra, this session — finite computation]
3. R-10: Draft honest reframe of Theorem 6.1. Identify which implication (“temperedness $\rightarrow$ GRH”) is the remaining conditional. Write the conditional version. [Lyra+Elie, this session]
4. Sarnak consultation: All three gaps would benefit from expert review. Update Sarnak letter with specific questions. [Keeper]

Analysis by Lyra, May 5, 2026. The elimination machinery (Sections 3-4) is sound pending R-9/R-11. The final inference (Section 6, Step 3) has a deeper issue. These are working notes — not final conclusions.